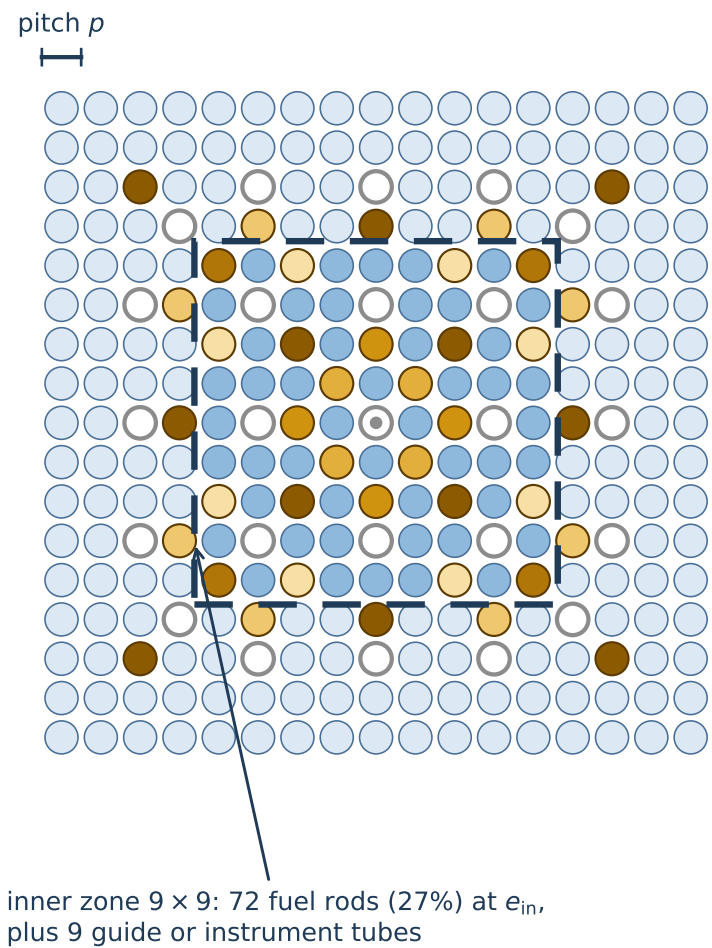


(a) 17 × 17 assembly
design variables set here: e_{in} , e_{out} , gd , n_{Gd} , p



Gd₂O₃ rod shading: the value of n_{Gd} at which the rod first appears (nested, octant-symmetric ladder)

● 12 ● 16 ● 20 ● 24 ● 32 ● 40 ○ guide tube ● e_{in} ● e_{out}

The six design variables

e_{in}	inner-zone enrichment	2.00 to 19.75	wt% ²³⁵ U	p	lattice pitch	1.15 to 1.43	cm
e_{out}	outer-zone enrichment	2.00 to 19.75	wt% ²³⁵ U	t_{refl}	reflector thickness	2.0 to 19.5	cm
gd	gadolinia weight fraction	0.0 to 8.0	wt% Gd ₂ O ₃	n_{Gd}	gadolinia rod count	12 to 40	rods, snapped to the ladder

264 fuel rods per assembly (72 inner and 192 outer), 24 guide tubes and 1 instrument tube. p and t_{refl} are coupled through g_{geom} , so the ranges above bound a box and not a feasible region.

(b) core plan view, drawn at $p = 1.26$ cm and $t_{refl} = 12$ cm
design variables set here: t_{refl} , p

